Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Enabled Autonomous Development of New Thermal Interface Materials (TIMs) for Advanced Semiconductor Packaging			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Devices and Systems	Other Electrical and Electronics Components	Fine chemicals	Precision Chemical Materials for the Electronics Industry
Concept of Proposed Technology	[Current Challenges & Market Needs] With the explosive growth of the AI computing market, 2.5D/3D Advanced Packaging and HBM (High Bandwidth Memory) have become standard. However, the thermal management is now the critical bottleneck limiting the maximum performance and manufacturing yield of these high-value devices. Existing TIMs (Thermal Interface Materials) fail to meet the extreme thermal density requirements and often cause reliability issues like pump-out or drying out under long-term operation. The market urgently demands a new class of TIMs that can deliver not only ultra-high thermal conductivity but also mechanical stability to prevent warpage and cracks in large-area packages. [Proposed Solution] Existing TIMs lack sufficient thermal conductivity, interfacial efficiency, and long-term durability under harsh conditions. Addressing these challenges requires not only better materials, but **a transformative approach to how new TIMs are designed, tested, and validated**. In this research, we aim to develop new AI Agents that can assist in TIM development workflows of simulation, prediction, and data-driven decision-making. For this, we plan to leverage cutting-edge AI technologies including Bayesian Optimization (BO) to guide formulation and design space exploration, Graph Neural Networks (GNN) and Physics-Informed Neural Networks (PINN)			

	for thermal transport prediction, and Reinforcement Learning (RL) for autonomous process optimization.
Rationale for Joint R&D Project	• The construction of robust AI architecture and the internalization of cutting-edge algorithmic capabilities require deep cooperation with world-leading AI research institutions. 1) Need for Global AI Technology Collaboration ○ Advanced AI techniques such as Graph Neural Networks (GNN), Physics-Informed Neural Networks (PINN), Bayesian Optimization (BO), and Reinforcement Learning (RL) are being rapidly advanced in AI-leading countries like Canada. The University of Toronto and the University of Waterloo host world-leading experts in AI and machine learning, supported by active research collaboration networks and strong interdisciplinary engagement through research centers such as Vector Institute(1) and Waterloo.AI(2) where the principle investigators (PIs) are affiliated. ○ International collaboration is essential to internalize these capabilities and advance Korea's own R&D capabilities in AI-driven materials innovation. 2) Strategic Response to U.S. Export Control Measures ○ Increasing US export control regulations pose significant challenges to international collaboration in AI and semiconductor technologies. ○ To secure technological sovereignty and ensure unrestricted access to critical AI capabilities, strategic partnerships with non-US AI powerhouses such as Canada are vital—both to diversify innovation pipelines and to reduce geopolitical vulnerability in Korea's high-tech supply chains.
Key Words	1. Thermal Interface Materials (TIMs) 2. Agentic AI 3. Materials Informatics 4.Closed-loop Process Control 5.Advanced Semiconductor Packaging

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	The ultimate goal of this research is to develop **next-generation AI Agents that can accelerate the development of advanced thermal interface materials**. Rather than fabricating specific TIMs directly, this research will focus on building an **AI-powered development architecture** that combines data-driven modeling, physics-based simulation, and automated optimization. Development of an Intelligent AI Agent Platform for TIM Innovation The main idea of this task is to build a modular AI Agent framework that integrates Graph Neural Networks (GNNs), Physics-Informed Neural Networks (PINNs), Reinforcement Learning (RL), and Bayesian Optimization (BO). This platform will autonomously manage the key stages of TIM development—design, simulation and parameter tuning—accelerating innovation while reducing trial-and-error cycles. Breakthrough in AI-Enabled Vertical Alignment and Electrical Insulation Design The AI Agents will be capable of proposing formulations and nanostructures that simultaneously achieve high thermal conductivity, electrical insulation, and vertical alignment—addressing key trade-offs in next-generation TIM development. Data-Driven Process Integration By combining simulation models and experimental data, the platform will enable iterative optimization of TIM design and processing conditions—adaptable to both lab-scale exploration and industrial-scale implementation.			
R&D Target Specifications	• The **AI Agents** will integrate state-of-the-art machine learning and simulation techniques for materials design, performance prediction, and process optimization to reduce development time and ensure robust performance across real-world applications. • New TIMs to be developed through this research aim to achieve target specifications such as 1) Thermal conductivity ≥ 15 W/m·K to effectively dissipate heat generated by high-power chips 2) Thermal resistance ≤ 0.1 K·cm²/W to eliminate interfacial thermal bottlenecks 3) Ensure compatibility with packaging processes to align with real-world manufacturing requirements such as viscosity, de-aeration, dispensability, and curing time 4) Ensure electrical insulation to prevent electrical conduction and enable broader applicability across diverse use cases • High Reliability Assurance for Advanced Packaging: To ensure long-term stability in 2.5D/3D packages, the developed TIM will specifically target 1) Adhesion Strength: Enhanced interfacial adhesion to prevent delamination 2) Stress Relaxation: Optimization of modulus to minimize Warpage and Crack formation 3) CTE Matching: Tuned Coefficient of Thermal Expansion to reduce thermal stress 4) Durability: Resistance to pump-out and dry-out under power cycling			
R&D Plan	• This study plans to apply cutting-edge AI techniques across all stages of the TIM development cycle—including composition design, interfacial engineering, process automation, and thermal simulation—via a unified, intelligent AI Agent platform. • Key Technical Components and Methods of Approach include - Thermal Performance Modeling Random Forest and GNN models will correlate multivariable data (e.g., filler type, loading, alignment, matrix properties) to thermal conductivity, identifying high-performance formulations early. - Interfacial Resistance Reduction GNN and PINN will be employed to simulate thermal pathways and quantify interfacial contact behavior using structural and chemical input parameters.			

	- Process Parameter Optimization Viscosity, curing temperature, and dispensing pressure will be optimized through ML/Bayesian models to ensure compatibility with industrial packaging workflows and maximize yield. - Thermal Simulation with PINN PINN will serve as a fast, accurate alternative to traditional FEM tools, enabling real-time simulation of heat flow and hotspot formation during iterative design

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• **AI Platform Development and Integration** Lead the architecture, development, and integration of the AI Agent platform, encompassing intelligent formulation, simulation, and experimental validation. Demonstrate the platform's capabilities in supporting material innovation, from initial design to real-world deployment. • **AI Agent-Based Formulation Design for High-Thermal-Conductivity Fillers** Perform AI-guided formulation and optimization of TIM compositions using Generative AI, machine learning (ML/DL), Bayesian Optimization (BO) and reinforcement learning (RL) for predictive and adaptive materials development. • **Thermal Analysis and Material Prediction Using GNN and PINN** Apply GNN and PINN to simulate and predict thermal conduction structures, process–property relationships, and reliability outcomes.
Expected R&R of Korean SME Partner	• **To identify and characterize cutting-edge ingredients potential to manage extreme thermal densities such as carbon-based composites, graphene-enhanced solutions, and synthetic diamond particles.** • **Compiling Large Data Sets of Component Materials, Product Performance Data, and Process Parameters** The research will require large amount of data sets to train various AI architectures across several product development stages. The Korean SME will lead the effort to - collect a large amount training datasets from the existing TIMs in production - provide experimental data across various combinations of nano-carbon fillers and polymers/oxides matrices, enabling composition-specific AI model training and design validation. • **Process Compatibility for High-Viscosity Composite Materials** Collaborate with the research institute to refine AI Agent-optimized TIM formulations and ensure compatibility with real-world semiconductor packaging processes (e.g., dispensing, curing, and de-aeration). • **Expansion of AI-Driven Intelligent Materials Design Platform** Extend the AI Agent framework developed for TIMs to additional applications such as battery thermal runaway prevention, high-thermal-conductivity plastics, and EMI shielding composites.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• **Expected technical outcomes** Development of an Intelligent AI Agent Platform for TIM Innovation If successful, this platform will autonomously manage the full cycle of TIM development—from design and simulation to process tuning and reliability prediction—significantly accelerating innovation while reducing trial-and-error cycles. Breakthrough in AI-Enabled Vertical Alignment and Electrical Insulation Design The AI Agent will be capable of proposing formulations and nanostructures that simultaneously achieve high thermal conductivity, electrical insulation, and vertical alignment—addressing key trade-offs in next-generation TIM development. • **Expected Industrial Impact** AI Platform as a Catalyst for TIM Localization and Commercialization The AI Agent will support domestic development of high-performance TIMs, which will lay a foundation for supply chain independence and Korean-led formulation IP. Cross-Sector Scalability of the AI-TIM Development Framework The platform will be applicable to broader high-heat applications such as AI semiconductors, electric vehicles, autonomous driving systems, telecom infrastructure, aerospace electronics, and data center cooling. Strengthening Korea's Semiconductor Ecosystem By facilitating co-optimization across materials, processes, and packaging requirements, the platform will support collaborative innovation and drive vertical integration across Korea's semiconductor ecosystem. • **Policy and Societal Contributions** Advancing the K-Semiconductor National Strategy through AI-Led Materials Design By internalizing core technologies for thermal management material design, the outcome of this research will directly contribute to Korea's competitiveness in system semiconductor and strategic material independence. Fostering AI-Materials Convergence Talent The research can serve as a platform for interdisciplinary talent development, fostering a new generation of experts in AI-driven materials science, reliability modeling, and advanced manufacturing.

6. References

(1) https://vectorinstitute.ai/

(2) https://uwaterloo.ca/artificial-intelligence-institute/

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization